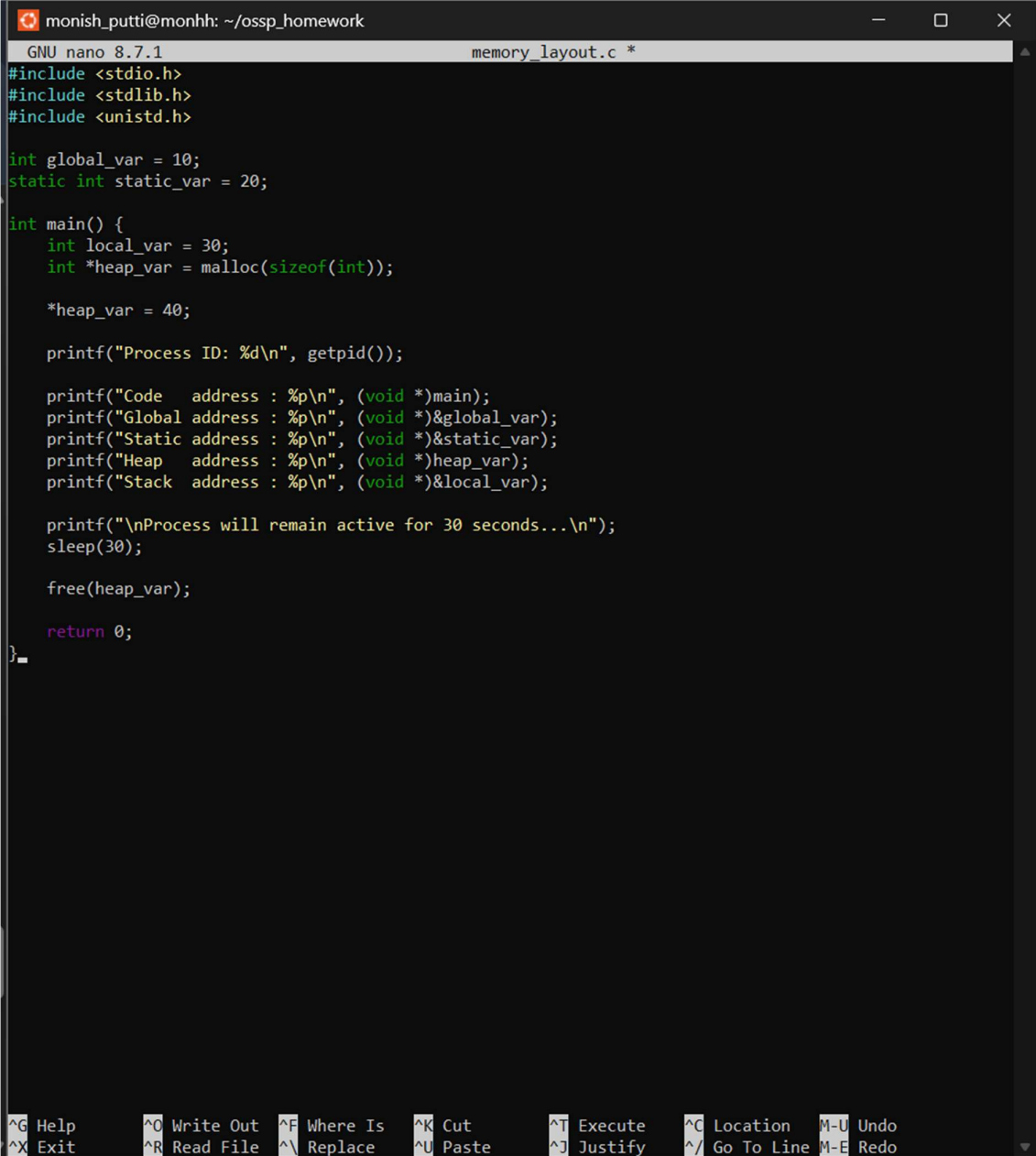


Ossp practical -7 and 8a , 8b

2520030395

Monish Putti

nano memory_layout.c



```
monish_putti@monhh: ~/ossp_homework
GNU nano 8.7.1 memory_layout.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

int main() {
    int local_var = 30;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 40;

    printf("Process ID: %d\n", getpid());

    printf("Code address : %p\n", (void *)main);
    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address : %p\n", (void *)heap_var);
    printf("Stack address : %p\n", (void *)&local_var);

    printf("\nProcess will remain active for 30 seconds...\n");
    sleep(30);

    free(heap_var);

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

```
monish_putti@monhh:/mnt/d/KLH_MONISH_395/Projects/KLH-CSE-2026-27-Team12-Process-Termination-Messages$ m
kdir -p ~/ossp_homework
cd ~/ossp_homework
monish_putti@monhh:~/ossp_homework$ nano memory_layout.c
monish_putti@monhh:~/ossp_homework$ gcc memory_layout.c -o memory_layout
monish_putti@monhh:~/ossp_homework$ ./memory_layout
Process ID: 936
Code   address : 0x631ad09f3209
Global address : 0x631ad09f6010
Static address : 0x631ad09f6014
Heap   address : 0x631aeb7a7010
Stack  address : 0x7ffd0af96fbc

Process will remain active for 30 seconds...
```

```
Process will remain active for 30 seconds...
monish_putti@monhh:~/ossp_homework$ cat /proc/12345/maps
cat: /proc/12345/maps: No such file or directory
monish_putti@monhh:~/ossp_homework$ nano memory_layout.c
monish_putti@monhh:~/ossp_homework$ gcc memory_layout.c -o memory_layout
monish_putti@monhh:~/ossp_homework$ ./memory_layout
Process ID: 988
Code   address : 0x57170a5d4209
Global address : 0x57170a5d7010
Static address : 0x57170a5d7014
Heap   address : 0x571744458010
Stack  address : 0x7fff0c2d987c

Process will remain active for 30 seconds...
```

nano memory_allocation.c

```
monish_putti@monhh: ~/ossp_homework
GNU nano 8.7.1 memory_allocation.c *
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a, *b;

    a = (int *)malloc(5 * sizeof(int));

    b = (int *)calloc(5, sizeof(int));

    a = (int *)realloc(a, 10 * sizeof(int));

    free(a);
    free(b);

    return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```

Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
monish_putti@monhh:~/ossp_homework$ valgrind --version
valgrind-3.26.0
monish_putti@monhh:~/ossp_homework$ valgrind --leak-check=full ./memory_allocation
==2281== Memcheck, a memory error detector
==2281== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2281== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2281== Command: ./memory_allocation
==2281==
==2281== HEAP SUMMARY:
==2281==     in use at exit: 0 bytes in 0 blocks
==2281==   total heap usage: 3 allocs, 3 frees, 80 bytes allocated
==2281==
==2281== All heap blocks were freed -- no leaks are possible
==2281==
==2281== For lists of detected and suppressed errors, rerun with: -s
==2281== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
monish_putti@monhh:~/ossp_homework$

```

8(b) — Copy-on-Write

```

monish_putti@monhh: ~/ossp_homework
GNU nano 8.7.1                                COW.C *
#include <stdio.h>
#include <unistd.h>

int main()
{
    int x = 100;

    if (fork() == 0)
    {
        printf("Child Before = %d\n", x);

        x = 500;

        printf("Child After = %d\n", x);
    }
    else
    {
        sleep(2);
        printf("Parent = %d\n", x);
    }

    return 0;
}

```

```
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
monish_putti@monhh:~/ossp_homework$ valgrind --version
valgrind-3.26.0
monish_putti@monhh:~/ossp_homework$ valgrind --leak-check=full ./memory_allocation
==2281== Memcheck, a memory error detector
==2281== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2281== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2281== Command: ./memory_allocation
==2281==
==2281==
==2281== HEAP SUMMARY:
==2281==    in use at exit: 0 bytes in 0 blocks
==2281==   total heap usage: 3 allocs, 3 frees, 80 bytes allocated
==2281==
==2281== All heap blocks were freed -- no leaks are possible
==2281==
==2281== For lists of detected and suppressed errors, rerun with: -s
==2281== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
monish_putti@monhh:~/ossp_homework$ nano cow.c
monish_putti@monhh:~/ossp_homework$ gcc cow.c -o cow
monish_putti@monhh:~/ossp_homework$ ./cow
Child Before = 100
Child After = 500
Parent = 100
monish_putti@monhh:~/ossp_homework$
```